Zamba: Computer vision for wildlife conservation

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DRIVENDATA

Data & AI solutions for problems that matter

We help mission-driven organizations harness their data to work smarter and offer more impactful services using data science, machine learning, and AI.

Data science & AI

- Machine learning model development
- Analysis & visualization
- Data science competitions

Software engineering

- Web application development
- Productionized machine learning

Data engineering

- Data pipelines
- Data warehousing

Data strategy

- Opportunity assessment
- State-of-the-field research

Agenda

- Wildlife conservation, camera traps, and the potential for machine learning
- What Zamba is and how it streamlines camera trap data analysis
- Key learnings in building impactful, practical machine learning tools for real-world use in conservation


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03-07-2018 09:42:48

Camera traps are widely used by researchers, conservationists, and park managers to monitor wildlife without human interference.

But they generate an **enormous amount of images and videos that need to be reviewed** and manually labeled by humans to identify specific footage of interest.

False triggers – caused by wind, rain, changes in light, etc. – are common, meaning **many capures do not contain an animal** at all, sometimes as much as 70%.

Machine learning is well-suited to this task. It has the potential to save countless hours of human review and enable better use of this valuable data.

Zamba

Zamba is a set of tools designed for wildlife conservationists to accelerate camera trap data analysis with machine learning and computer vision.

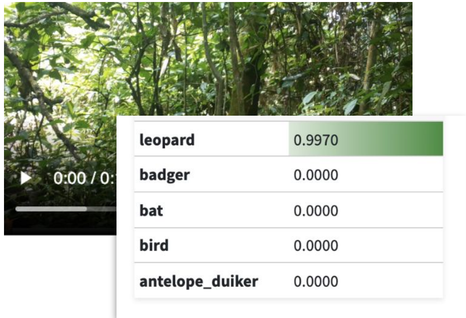
In building and supporting Zamba, we found that an effective solution needs to be **accurate**, **extendable**, and **accessible**.

Developing
accurate models

Zamba's modeling tasks

Species classification

Predict species in camera trap videos or images using models trained on large, expert-labeled datasets



→ Filter blank inputs and select specific animal populations

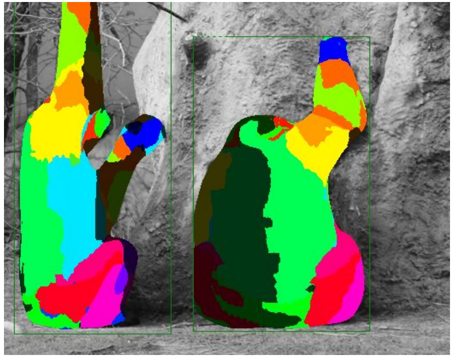
Depth estimation

Predict distance at 1 frame per second for bushbucks, chimpanzees, duikers, elephants, leopards, monkeys

→ Estimate population size using statistical methods

Anatomical pose

Dense segmentation and map to specific anatomy for chimpanzees



→ Study animal behavior

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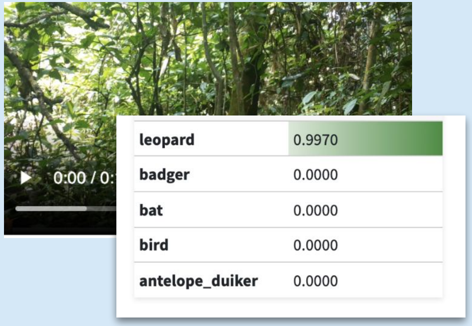

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per secon
chimpanz
elephants

The focus of today's talk

→ Estimate population size using statistical methods

Anatomical pose

Dense segmentation and map to specific anatomy for chimpanzees

→ Study animal behavior

Video species classification

Video pipeline

Videos contain a lot of data

- Often 30 FPS and 1 minute long
- Processing every frame is impractical

Animals are often present only in a minority of frames

One of the main challenges in video classification is **frame selection**

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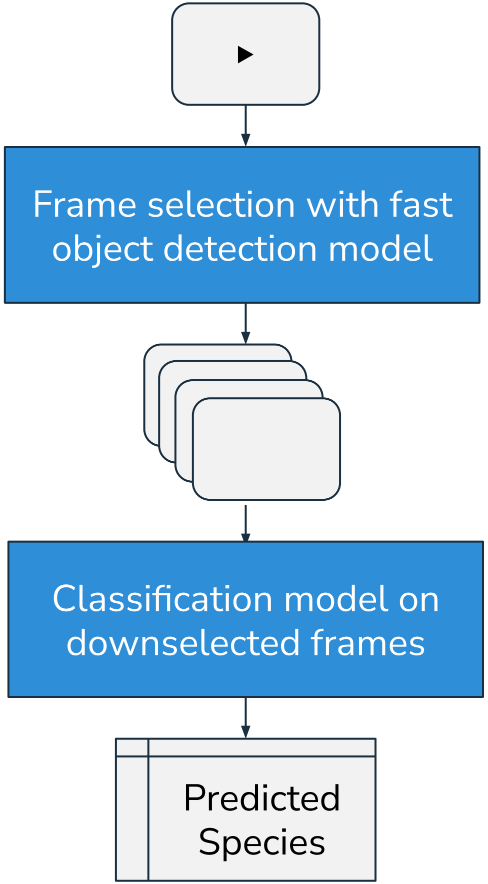

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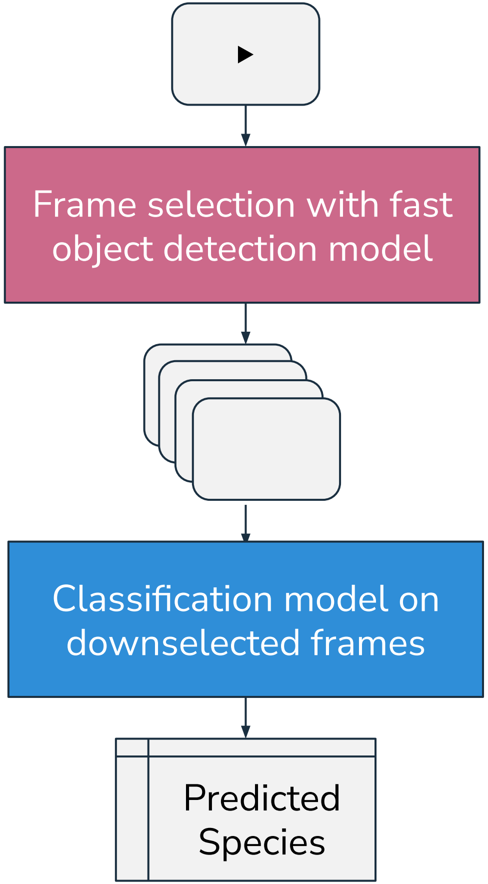
Frame selection needs to be

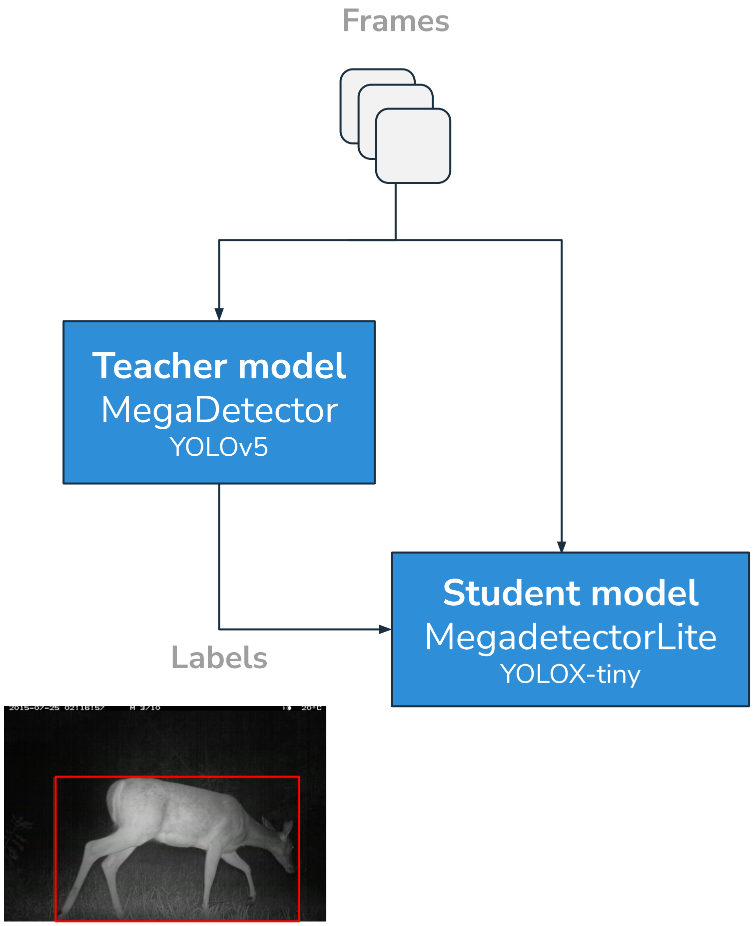
- Fast (upstream task)
- Reasonably accurate (selecting blank frames prohibits accurate species classification)



Frame selection

Zamba's default "smart" frame selection uses a fast object detection model called **MegadetectorLite**

Trained with knowledge distillation technique on labels from MegaDetector, an accurate but computationally intensive species-agnostic object detection model for camera trap images



Species classification

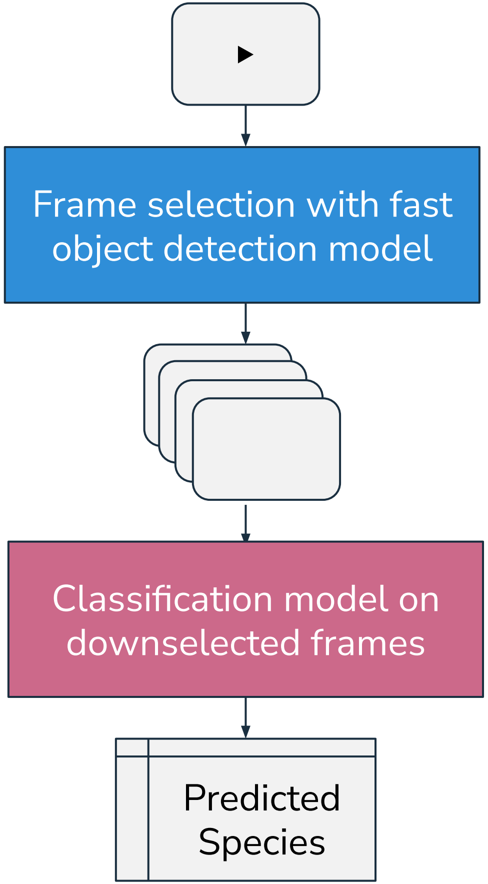
Zamba provides two different model architectures for species classification on videos

Image-native model

- Faster to train and predict
- Simpler architecture to fine-tune

Video-native model

- More accurate for smaller animals
- Able to pick up on cues from motion



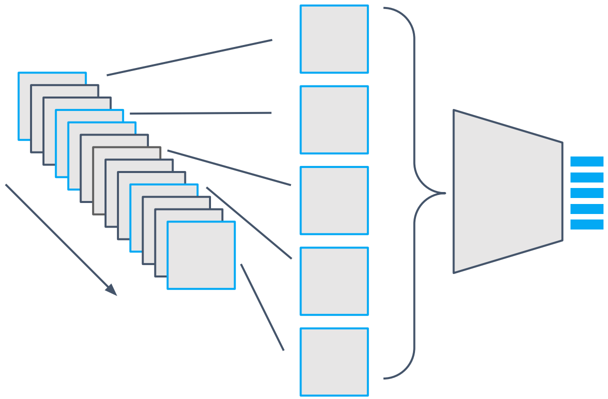
Can you identify the animal present?

How about now?

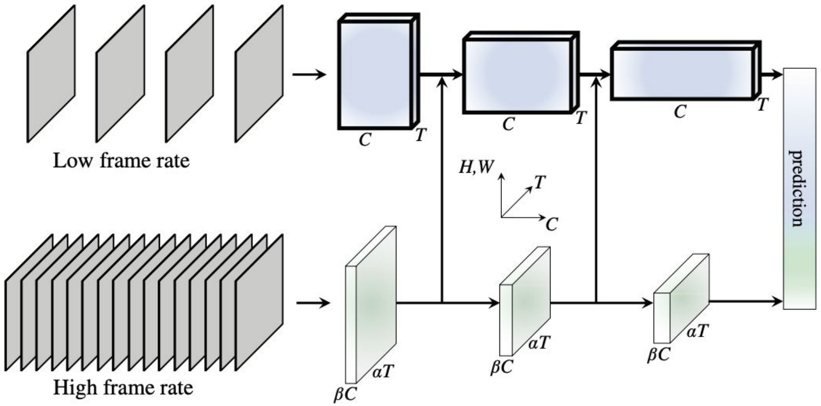
This slide showed a video where there is visible motion of a bird flying away.

Classification model architectures

Time-distributed EfficientNet
(Image-native)



Slowfast
(Video-native)



(Feichtenhofer et al., 2018) "SlowFast Networks for Video Recognition"

Processes frames as images and then aggregates into a single prediction.

Slow pathway captures spatial semantics and fast pathway captures motion.

Species classification models

Training data

The video models were trained on 250,000 labeled camera trap videos from camera trap deployments across 14 countries

Half of this data was annotated by trained ecologists, while the other half consists of high-confidence labels generated by citizen scientists through the Chimp&See platform

Pre-trained models

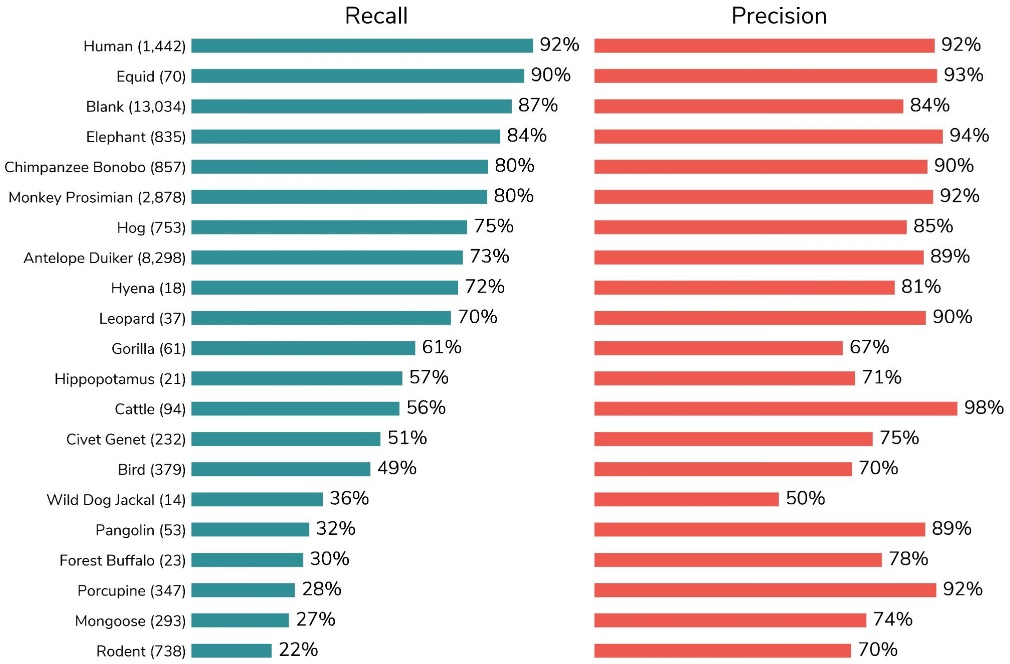
- **African forest species**
  32 forest species commonly found in Central and West Africa
- **European species**
  11 species commonly found in Western Europe
- **Blank detection**
  Binary classification model for detecting videos with no animal

African forest species model performance

Overall top-1 accuracy: 82%
Overall top-3 accuracy: 94%

Model performance is generally worse for smaller species, less common species, heavily occluded species, and nocturnal species

As camera trap deployments vary greatly, users should always evaluate model performance on their own data



Performance metrics for African forest species model on a holdout set of 30,324 videos


Bushnell
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Image species classification

Image support in Zamba

The vast majority of camera trap users collect images because they are easier to collect, store, and process.

There are existing applications that provide pretrained species models for camera trap image data.

To differentiate our contribution, Zamba's image support emphasizes **training custom models**.

Zamba has one pretrained image model that is intended as a **generalizable base model** for fine-tuning.

Wildlife Insights

Microsoft | AI for Good Lab
Pytorch Wildlife

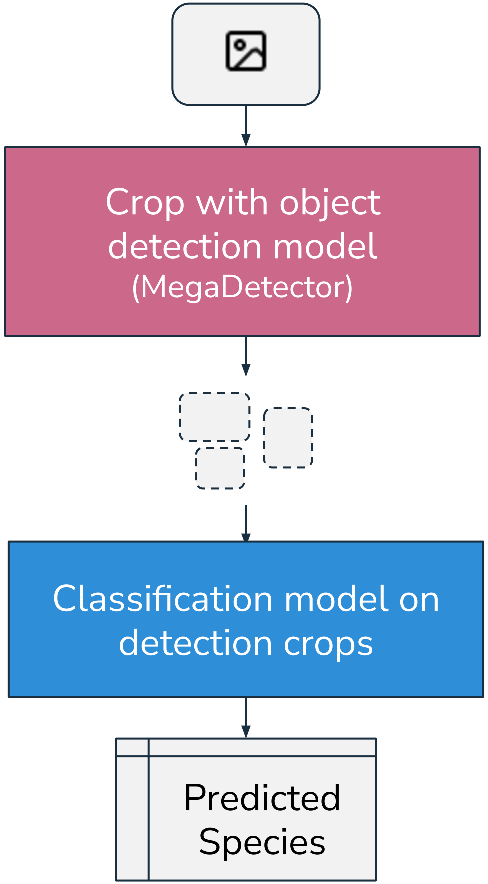
AddaxAI

Image pipeline

Instead of frame selection, the first preprocessing step is to **detect and crop to the animal**

- Gadot et. al (2024) found that classification performance improves significantly (25% improvement on macro F1) when predicting on crops of the animal rather than the full image

Since images are faster to process than videos, we use MegaDetector, the species-agnostic object detection teacher model discussed earlier



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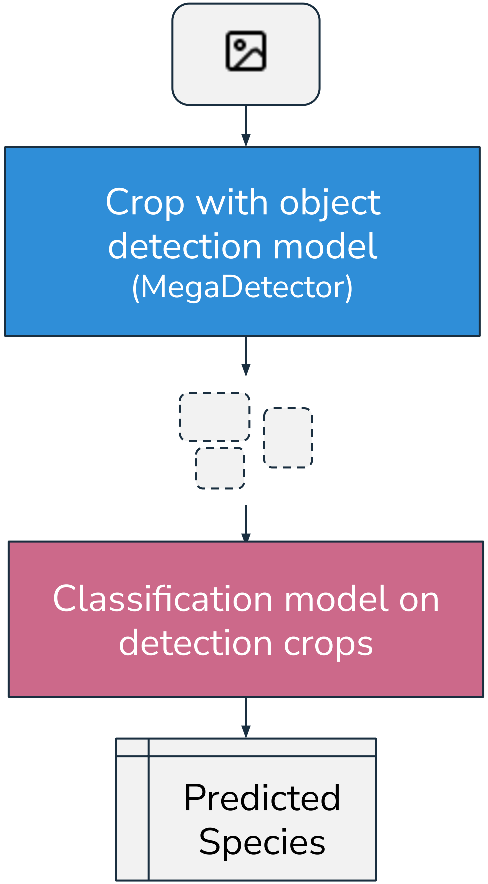
Classification model architecture

Zamba's image model uses **ConvNext V2 Base** as its backbone



Classification model architecture

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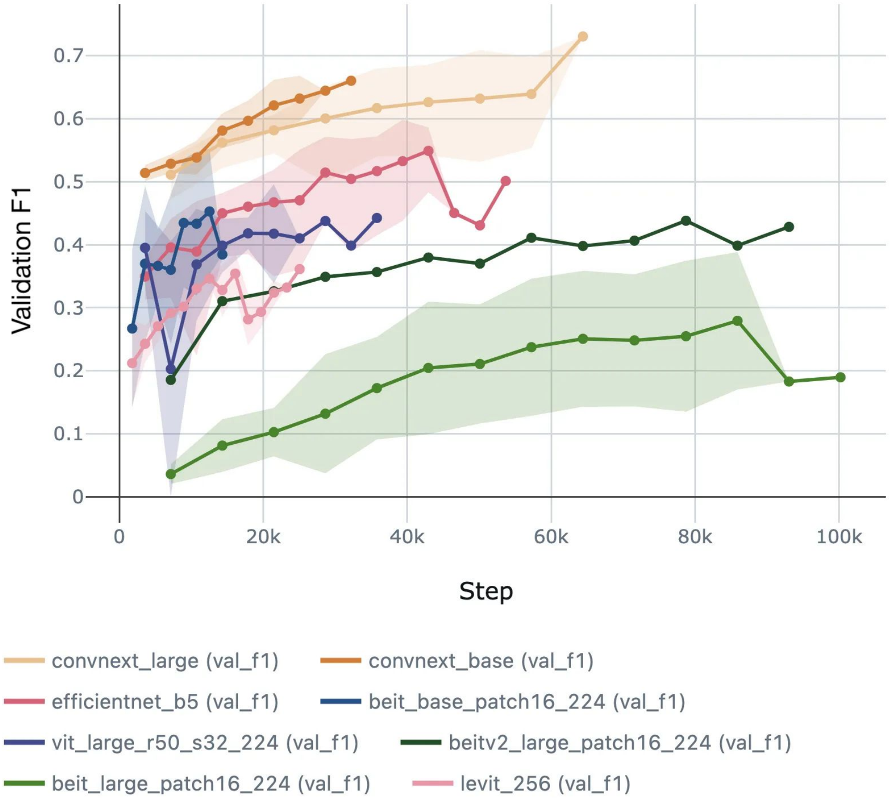
ConvNext variants were found to consistently achieve higher accuracy with fewer training epochs in our experiments



Species classification model

Training data

The image model was trained on over 15 million annotations from over 7 million images

It was aggregated from 20 terrestrial camera datasets from LILA BC, representing a wide range of global habitats

To standardize labels, we reviewed 1,231 label classes across the source datasets and curated a unified taxonomy of 178 label classes

Pre-trained model

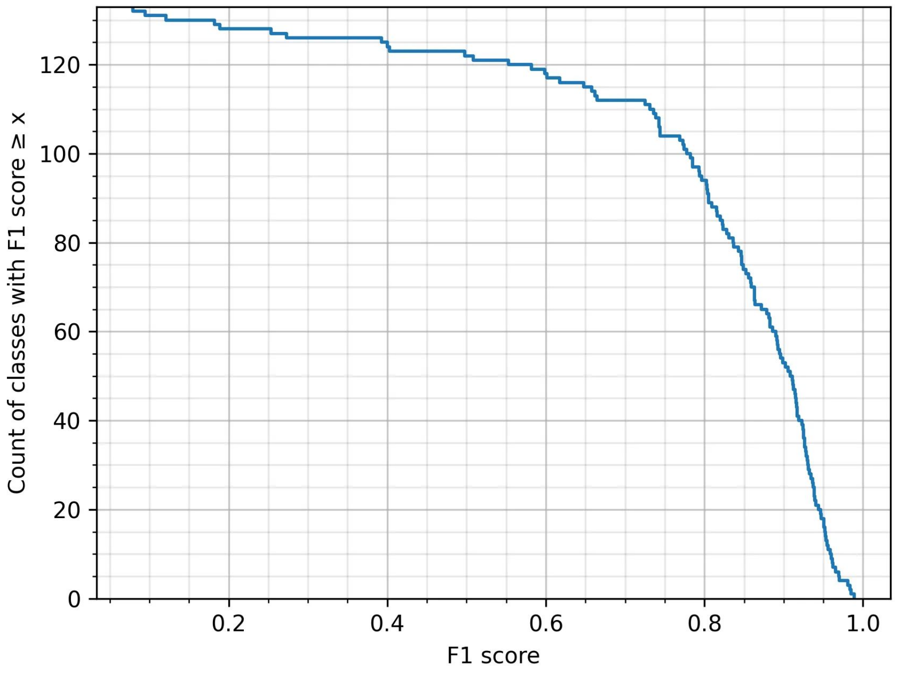
- **Global species base model**
  178 species and species groups

Global species base model performance

Overall top-1 accuracy: 90%

On a holdout set of ~450,000 annotations, 40% (53 classes) have an F1 score of over 0.9 and 71% (94) have an F1 score of over 0.8

As with videos, model performance is heavily dataset- and context-dependent and users should always evaluate model performance on their own data



The curve shows the count of label classes whose F1 score exceeds the value given by the x-axis. Only classes with >=30 observations are included.

Developing

extendable models

The need for custom models

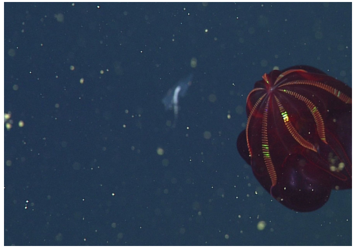
Camera trap applications vary greatly

- Habitat
- Object of study
- Taxonomic rank

Pretrained models are insufficient to capture the diversity of conservation workflows

It's critical to support conservationists in training custom models designed specifically for their use case





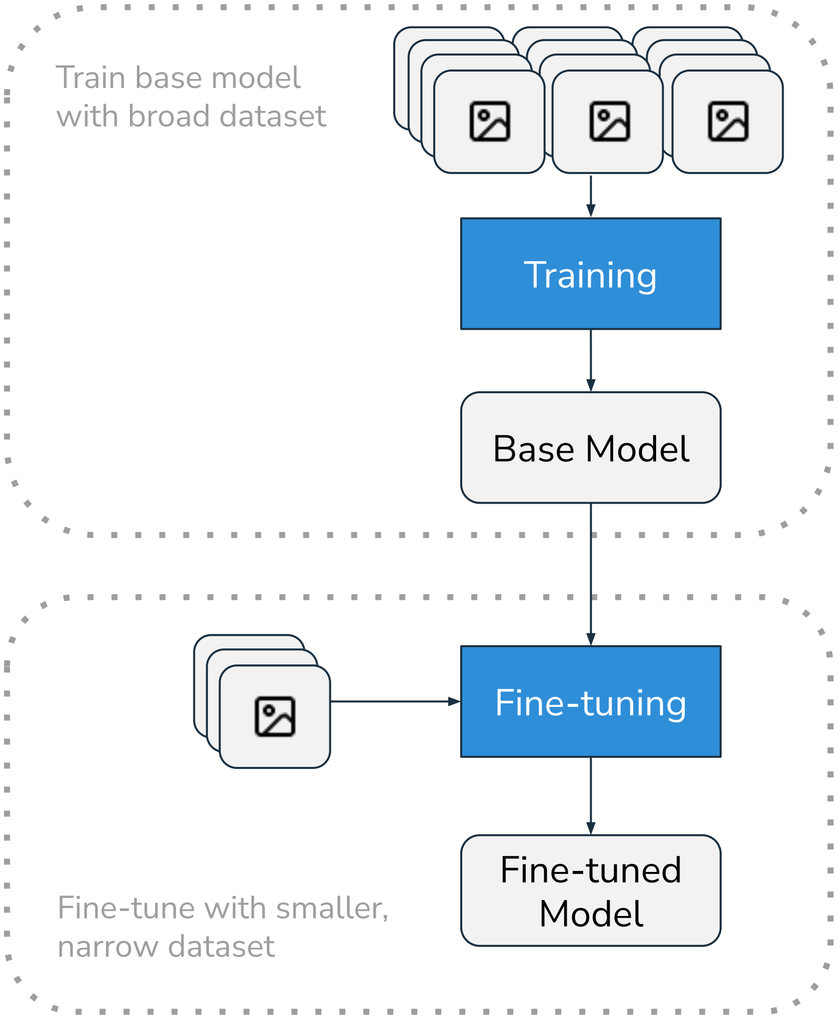



Model fine-tuning

Zamba includes a first-class fine-tuning workflow for both videos and images

The pretrained models are designed with robust geographical and species representation so that they are capable base models

Through experimentation, we have determined effective reasonable defaults for hyperparameters, such as splits, learning rate, and early stopping



Model fine-tuning

Fine-tuning can be highly effective with as little as a few hundred examples per label class

This fits well with the circumstances of many conservation projects, which have data from prior manual annotation

All you need is to bring the media files and a CSV with file paths and labels

	filepath	label
0	raw/passerines_etc/dunnock/LIFT0079.MP4	dunnock
1	raw/passerines_etc/dunnock/07220261.MP4	dunnock
2	raw/passerines_etc/dunnock/LIFT0075.MP4	dunnock
3	raw/passerines_etc/dunnock/LIFT0097.MP4	dunnock
4	raw/passerines_etc/dunnock/LIFT0078.MP4	dunnock
...	...	...
3051	raw/takahe/IMAG0058.AVI	takahe
3052	raw/takahe/IMAG0004 (6).AVI	takahe
3053	raw/takahe/IMAG0086 (2)-1.AVI	takahe
3054	raw/takahe/IMAG0084.AVI	takahe
3055	raw/weasel/SOIK_LIFT0037 - Copy.AVI	weasel

3056 rows × 2 columns

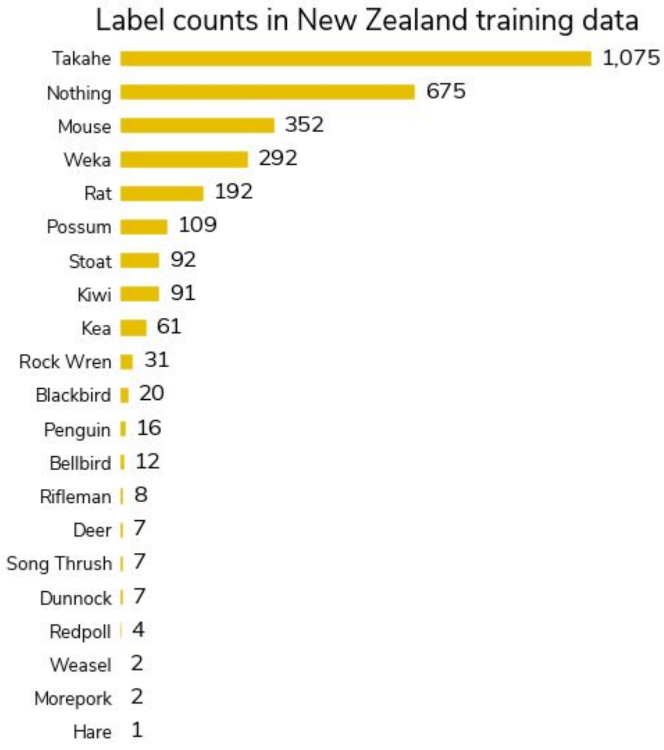
Example labels CSV for fine-tuning workflow

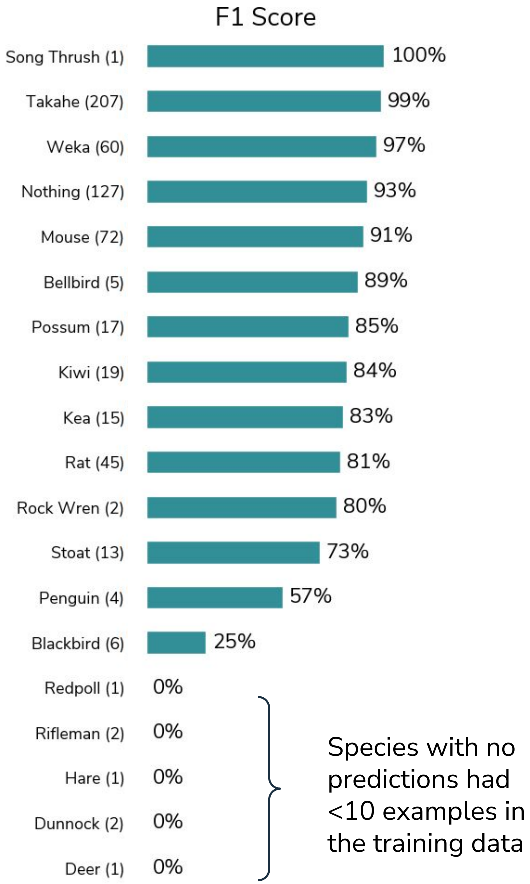
Fine-tuning example

Fine-tuned video species classification model on an entirely new set of species in a new ecology: New Zealand birds and rodents

Training dataset has 3,056 videos

Fine-tuned model achieves **91% top-1 accuracy** on holdout set of 600 videos





Developing **accessible** models

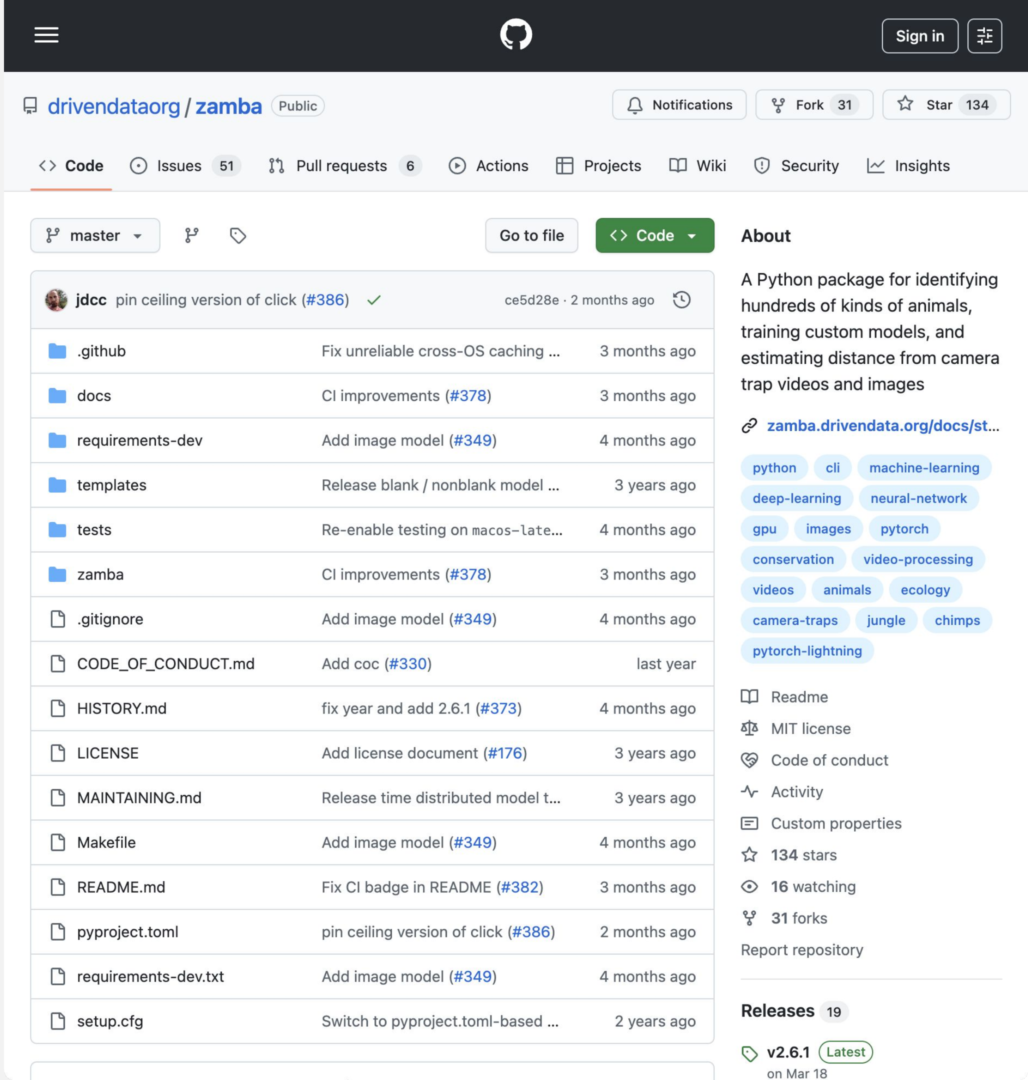
Zamba is an open source Python package

Zamba is built on top of a popular open source deep learning (PyTorch) and other pieces of the scientific Python ecosystem

It is itself an **open source package**

Being open source:

- Invites collaboration
- Fosters transparency and trust
- Promotes integration and portability



Zamba is easy to use

Access functionality through a simple command-line interface (CLI)

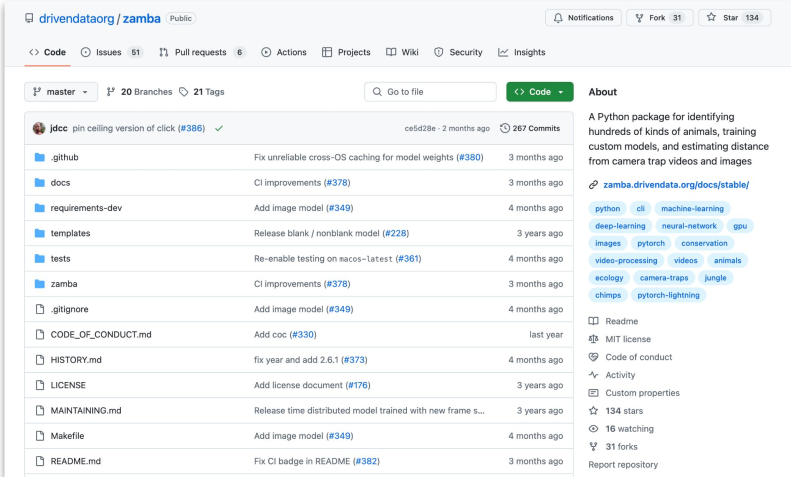
```
# Predict
zamba predict --data-dir path/to/videos


# Fine-tune
zamba train --data-dir path/to/videos \
  --labels my_labels.csv
```

But many conservationists are not programmers.

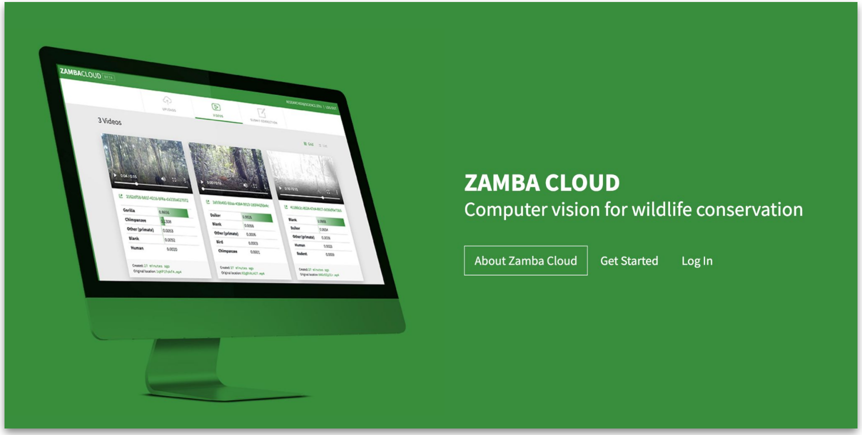
Zamba still needs to be accessible for these users.

Zamba powers Zamba Cloud, a code-free interface

Zamba



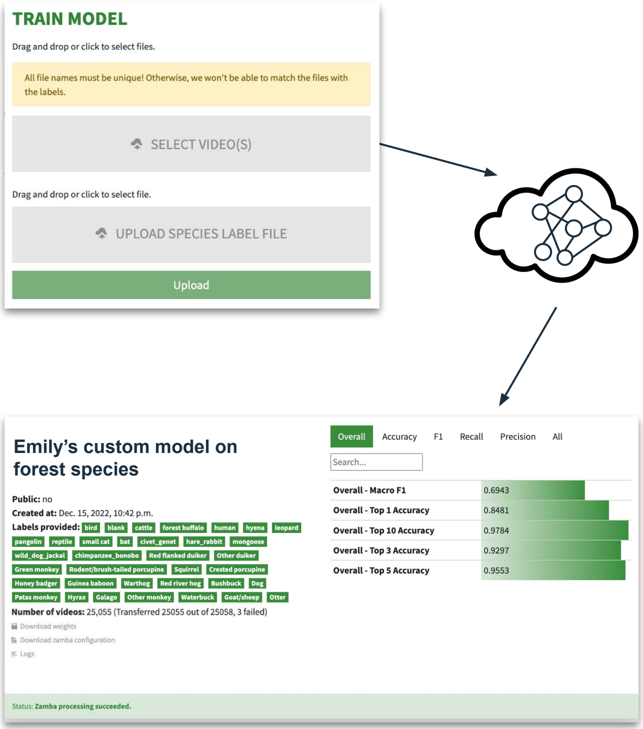
Zamba Cloud



Use all of the functionality through a simple CLI, or inspect, extend, or contribute to the open code base.

Predict with the pretrained Zamba models or fine-tune custom models — without writing any code — just by uploading data.

Zamba Cloud: a code-free way to train and run ML models



blank	0.9812
monkey_prosimian	0.0029
bird	0.0019
mongoose	0.0010

antelope_duiker	1.0000
badger	0.0000
bat	0.0000
bird	0.0000

elephant	0.5085
gorilla	0.1462
hog	0.0470
monkey_prosimian	0.0311

Model fine-tuning

Model inference

Download the predictions or model weights

The **CSV output format for predictions** is easy to use for analysis with Excel or other any statistical software. The open format is simple and portable.

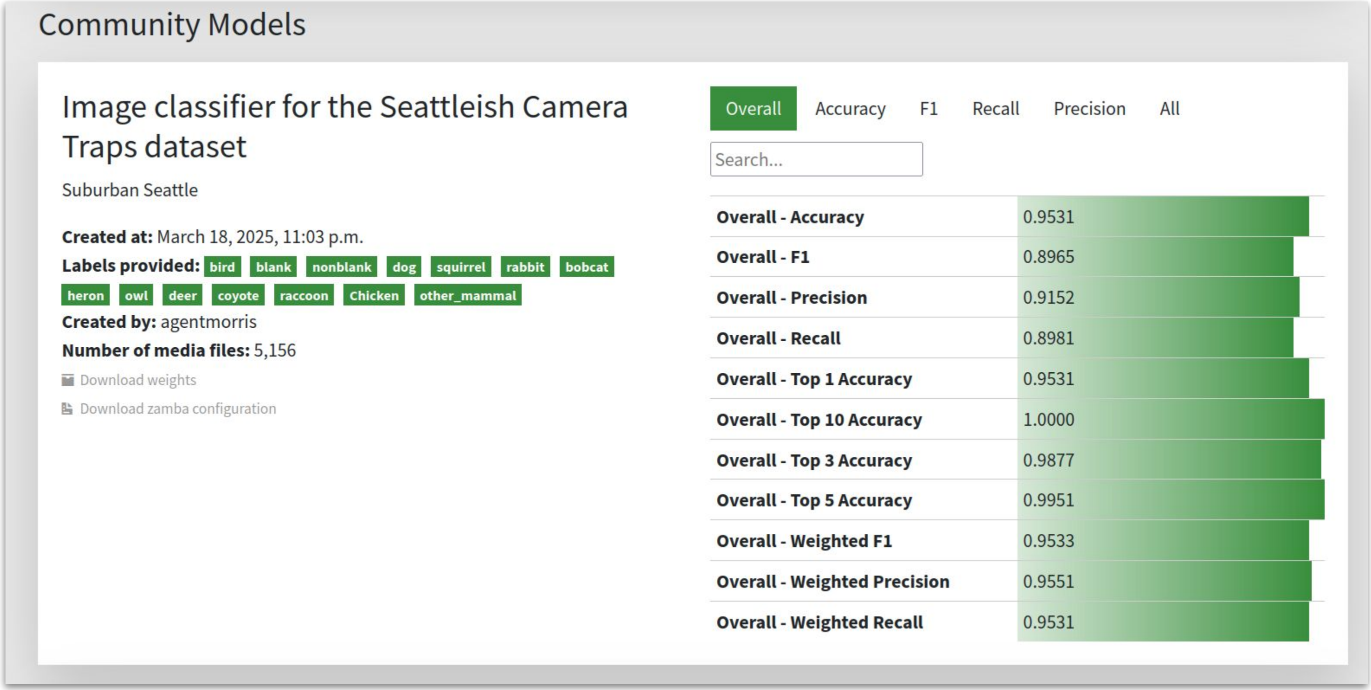
The **PyTorch model weights** can be downloaded and used with the open source Zamba package.

Media ID	MegaDetector outputs		Bounding box				Individual species probabilities					
filepath	detection category	detection conf	x1	y1	x2	y2	american black bear	american crow	bird	bobcat	chicken	c
image_0.jpg	1	0.6240	253	295	375	353	0.0887	0.3027	0.0327	0.0148	0.0366	0
image_1.jpg	1	0.9140	211	90	352	258	0.0895	0.1190	0.0752	0.0144	0.0232	0
image_2.jpg	1	0.9410	547	281	694	389	0.0001	0.0001	0.0001	0.0003	0.0000	0

Example predictions for image species classification

Since 2018, Zamba has served over 300 users from around the world. We've processed more than 1.1 million videos.

Our image support is new since March this year, and we're excited to see people using it.



Community Models

Image classifier for the Seattleish Camera Traps dataset

Suburban Seattle

Created at: March 18, 2025, 11:03 p.m.
Labels provided: bird blank nonblank dog squirrel rabbit bobcat heron owl deer coyote raccoon Chicken other_mammal
Created by: agentmorris
Number of media files: 5,156

Download weights

Download zamba configuration

Overall Accuracy F1 Recall Precision All

Search...

Overall - Accuracy	0.9531
Overall - F1	0.8965
Overall - Precision	0.9152
Overall - Recall	0.8981
Overall - Top 1 Accuracy	0.9531
Overall - Top 10 Accuracy	1.0000
Overall - Top 3 Accuracy	0.9877
Overall - Top 5 Accuracy	0.9951
Overall - Weighted F1	0.9533
Overall - Weighted Precision	0.9551
Overall - Weighted Recall	0.9531

By enabling conservationists to efficiently analyze large datasets and train models tailored to their specific ecological contexts, Zamba advances wildlife monitoring, research, and evidence-based decision-making.

Key takeaways

- **Accurate** machine learning models can effectively identify species of interest in camera trap videos and images
- **Extendable** models enable Zamba to be useful to everyone who collects camera trap data
- **Accessibility** is critical for actually getting machine learning into the hands of conservationists

Zamba is the work of many people

Emily Dorne

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Justin Chung Clark

DATA SCIENTIST

Robert Gibboni

SENIOR DATA SCIENTIST

Katie Wetstone

SENIOR DATA SCIENTIST

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WILDLABS.NET [The conservation technology network]

arcus FOUNDATION

Patrick J McGovern FOUNDATION


Bushnell
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07-22-2019 07:06:44

Thank you!
Questions?

www.zambacloud.com

github.com/drivendataorg/zamba

DRIVENDATA